

Phishing Between Loneliness and Skepticism: How Social Engineering Exploits the Need for Belonging, Safety, and Control

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Abstract (draft)

Phishing remains one of the most common entry points to serious cyber incidents, despite rising public awareness and technical controls. This thesis investigates why phishing continues to work in a context characterised by both increased loneliness and increased general skepticism toward strangers online. The core argument is that phishing often does not resemble a stranger attack. Instead, it imitates what people still long for or fear losing: belonging, safety, order, status, help, and control.

Using a mixed-methods design among adults in higher education in Norway, the study examines emotional and contextual vulnerability markers, the gap between knowing about phishing and acting safely under pressure, and which protective factors may reduce risk in practice. The expected contribution is a human-centred explanation of phishing susceptibility that moves beyond the "weakest link" narrative and informs better awareness, culture, and design.

Keywords: phishing, loneliness, social engineering, human factors, protection motivation, cybersecurity behaviour

1. Introduction

Cybersecurity is often presented as a technical problem with a human weak point. This thesis starts from a different premise: phishing is a human decision problem disguised as an email.

Every year, organisations invest in filters, training, policies, and incident response. Yet phishing and related social engineering attacks remain among the most effective ways to gain initial access to systems, accounts, and sensitive information (ENISA, 2024; CISA, n.d.). The problem persists not only because attackers improve their tools, but because they improve their understanding of human motivation.

In Norway, public awareness of digital fraud has increased. People are told to be skeptical of unknown senders, unexpected links, and requests for credentials. At the same time, loneliness and social disconnection are widely discussed as social problems across age groups (ONS, 2023; Nicholson et al., 2019). This creates a tension that is central to the thesis:

In a time of both rising loneliness and rising general skepticism, phishing becomes dangerous precisely because it does not look like an attack from a stranger. It imitates what people still long for or fear losing: belonging, safety, order, status, help, and control.

The thesis therefore asks not only why people click, but why emotionally and socially meaningful lures can bypass skepticism that would otherwise protect them from obvious fraud.

1.1 Background and relevance

Recent Norwegian warnings from Finans Norge, banks, NSM, and Nkom show that social engineering remains a live threat. Attacks combine email, SMS, phone calls, impersonation, urgency, and authority. AI-assisted phishing may increase credibility and scale, but the underlying mechanism remains psychological: influence under uncertainty (Halcyon, 2026; Trend Micro, 2026).

For a Norwegian master thesis in cybersecurity, this topic is timely and feasible. It connects technical security with human behaviour, privacy, trust, and social context - without requiring access to classified infrastructure data.

1.2 Research problem

Main research question:

How can phishing remain effective in a context of increased loneliness and increased general skepticism, and which emotional, cognitive, and contextual factors make individuals especially vulnerable to this form of social engineering?

Sub-questions:

1. How do loneliness, social connection needs, and general online skepticism relate to phishing susceptibility?
2. Which emotional and situational factors (urgency, authority, fear, shame, stress) most strongly influence unsafe responses?
3. What is the gap between knowing about phishing and acting safely in the moment of decision?
4. Which protective factors - training, culture, verification routines, peer support - are experienced as most effective?

1.3 Scope

- **In scope:** phishing, smishing, vishing, impersonation scams, emotional and social manipulation online.
- **Population:** adults in higher education in Norway, primarily 18-35 years.
- **Out of scope as main focus:** malware reverse engineering, DDoS, ransomware operations, deep technical forensics.
- **Geographic frame:** Norway, with international literature as theoretical backdrop.

1.4 Contribution

The thesis aims to contribute:

- a human-centred model of phishing vulnerability beyond the "stupid user" stereotype;
- empirical insight into loneliness, skepticism, and decision-making under manipulation;
- practical recommendations for awareness, education, and security culture in universities and early-career workplaces.

2. Background: Phishing as an Entry Point

2.1 What is phishing?

Phishing is a social engineering attack in which the victim is manipulated into clicking a malicious link, opening an attachment, revealing credentials, approving an action, or transferring money. The attack succeeds when the message appears credible, relevant, or emotionally meaningful (Hadrnag, 2018; CISA, n.d.).

2.2 Phishing as the starting point in attack chains

Many serious incidents begin with a human action: a click, a reply, a login, or an approval. From there, attackers may pursue account takeover, data theft, business email compromise, or further exploitation. This makes phishing analytically important even when the final harm is ransomware or fraud.

2.3 Norway and digital trust

Norway is highly digitalised. Citizens rely on BankID, ID-porten, Altinn, digital mailboxes, and online banking. High trust in institutions can be a societal strength, but it may also create fertile ground for impersonation attacks when combined with urgency and fear. Public campaigns such as Finans Norge's "self-defence against fraud" (selvforsvar mot svindel) show that awareness exists - yet incidents continue.

2.4 AI and the new scale

AI can improve grammar, personalisation, translation, and impersonation. That lowers the cost of credible phishing, but it does not replace the need for emotional leverage. AI makes the lure easier to produce; human needs determine whether it works.

2.5 Norwegian context: loneliness, students, and phishing

Norway combines high digital trust with measurable social disconnection in key groups. SSB's Livskvalitetsundersøkelse (2025) finds that 14.5% of adults report being quite or very troubled by loneliness in the last two weeks, with the highest rates among 18-24-year-olds (23.3%) and 25-44-year-olds (19.1%) (Statistics Norway, 2025). Among students specifically, a 2024 NSO/Khrono survey of 1,000 students found that 68% report some degree of loneliness, including 22% who feel lonely to a large or very large extent (NSO & Khrono, 2024).

At the same time, phishing remains the dominant form of digital fraud in Norway. NSM received more than 100,000 reports of phishing attempts in 2024, and the 2026 risk report emphasises that social manipulation attacks are increasing in sophistication, including AI-assisted content and attacks that bypass traditional MFA (NSM, 2026). This juxtaposition -

high loneliness among young adults and students, plus rising phishing volume - makes the thesis question locally relevant, not only theoretically interesting.

Indicator	Source	Relevance to thesis
14.5% adults quite/very lonely (2025)	SSB Livskvalitet	Baseline population context
23.3% of 18-24 report strong loneliness	SSB	Core age group in HE sample
68% students report some loneliness	NSO/Khrono 2024	OsloMet/HE recruitment frame
100,000+ phishing reports (2024)	NSM	Threat is live and measurable
AI-assisted phishing rising	NSM Risiko 2026	Human factors remain decisive

3. Literature Review

3.1 Phishing and human factors

Research consistently shows that phishing success depends on persuasion strategies, context, workload, and decision pressure rather than intelligence alone (Parker & Flowerday, 2020; Tsai et al., 2021). Employees and students may know the rules yet still click when messages appear legitimate, urgent, or socially relevant.

3.2 Loneliness and fraud

Loneliness has long been associated with fraud vulnerability among older adults (Alves & Wilson, 2008; Wen et al., 2022). Recent experimental work suggests loneliness can increase susceptibility to misleading offers in middle-aged and older adults (Wen et al., 2024). Mechanisms include increased desire for connection, greater trust in persuasive strangers, and reduced access to social verification.

However, the relationship is not uniform across age groups. Some studies find weaker or more complex effects among younger adults, where FOMO, social media exposure, impulsivity, and platform-specific cues may matter more (Przybylski et al., 2013; Popovac & Flowerday, 2024). Among college students, loneliness and impulsivity have been linked to gullibility through chain mediation (Nature HSMC, 2025).

Research gap: loneliness is well studied in older fraud victims, but less systematically examined in Norwegian higher-education contexts and in relation to the skepticism paradox.

For the OsloMet sample, student loneliness is not marginal. SHoT 2022 found that 29% of students often miss someone to be with, and campus presence correlates with lower loneliness (SHoT, 2022). OsloMet initiatives such as "Lunch Friend" (Lunsjvenn) project (matching students for lunch) confirm that institutions recognise social belonging as a student-welfare issue (OsloMet, 2024). This thesis connects that welfare concern to cybersecurity: the same need for connection that drives welfare interventions may also shape responses to socially framed phishing.

3.3 Skepticism without protection

People may report high skepticism toward unknown senders yet remain vulnerable to messages that appear official, socially relevant, or emotionally important. Young adults are highly exposed on social platforms and may respond differently to phishing that mimics peers, events, or opportunities rather than strangers (Popovac & Flowerday, 2024). Gen Z interview research also shows awareness of risk without confidence in how to respond or seek support (AMCIS, 2025).

3.4 Protection Motivation Theory (PMT)

PMT explains protective behaviour through threat appraisal (severity and vulnerability) and coping appraisal (response efficacy and self-efficacy) (Rogers, 1975; Floyd et al., 2000). In cybersecurity, PMT has been used to explain reporting behaviour and resistance to phishing (Tsai et al., 2021; Bayl-Smith et al., 2024).

For this thesis, PMT is useful but incomplete. It explains motivation to protect oneself, yet phishing often succeeds when the message does not initially look like a threat. Emotional needs may reframe the lure as an opportunity, warning, or social contact rather than danger.

3.5 Remote work, cognitive bias, and shame

Remote and screen-mediated communication reduces social cues available for verification. Research on remote workers shows that isolation can increase susceptibility to social engineering that exploits trust and the desire for connection (Journal of Technology and Science, 2024). Overconfidence bias may coexist with loneliness: people may believe they are "too smart" for phishing while remaining vulnerable to emotionally relevant lures.

Shame and fear of judgment reduce reporting. Training that relies on simulated phishing without support-seeking components may increase awareness but not safe behaviour under stress (Chen et al., 2024). For lonely individuals, romance and confidence scams show that victims may return to contact with scammers because the relationship itself feels meaningful (WIRED, 2024). Universities need non-punitive reporting cultures if data collection and prevention are to reflect real behaviour.

3.6 Theoretical synthesis

Perspective	Contributes	Limitation
Social engineering / persuasion	Authority, urgency, fear, liking, scarcity	Can underplay emotional loneliness
Loneliness research	Connection needs, social void, trust in strangers	Often focused on older adults
PMT	Threat and coping appraisals	Assumes threat is recognised as such
Security culture	Reporting, verification, non-punitive learning	Needs organisational context

The thesis proposes a combined model: **phishing susceptibility = emotional relevance + situational pressure + low opportunity for verification - perceived social reward.**

4. Theoretical Framework

4.1 Core hypothesis

Phishing succeeds when it imitates legitimate human and institutional needs more convincingly than general skepticism can counter in the moment of decision.

4.2 What phishing imitates

Need / fear	Example lure	Why it may bypass skepticism
Belonging	Message from "colleague", romance scam, event invite	Feels socially relevant, not random
Safety	"Your account is compromised"	Triggers protective action
Order / control	Tax, invoice, admin deadline	Promises to fix a problem quickly
Status	Job offer, exclusive opportunity	Appeals to identity and ambition
Help	IT support, "we can fix this for you"	Uses helpfulness and compliance

4.3 Hypotheses (working version)

1. H1: Higher loneliness is associated with higher self-reported phishing susceptibility, especially for socially framed lures.
2. H2: Higher general skepticism alone does not predict lower susceptibility if the lure appears emotionally or institutionally relevant.
3. H3: Situational pressure (urgency, authority, stress) mediates the relationship between loneliness and unsafe intended behaviour.
4. H4: Participants with stronger verification habits and supportive security culture report lower susceptibility.

Age will be treated as a control/exploratory variable because prior research suggests loneliness may operate differently across cohorts (Wen et al., 2024).

5. Methodology

5.1 Design

Recommended design: exploratory sequential mixed methods.

1. Phase 1: Online survey (quantitative + short open text)
2. Phase 2: Semi-structured interviews with selected respondents
3. Phase 3: Thematic analysis integrated with survey findings

5.2 Population and sample

- **Population:** students and staff in higher education in Norway.
- **Target sample:** 120-180 survey responses; 12-18 interviews.
- **Inclusion:** adults 18+ with regular digital communication for study/work.
- **Recruitment:** OsloMet programmes, student networks, LinkedIn, privacy-preserving invitation text.

5.3 Survey instruments (draft)

Construct	Instrument / measure	Notes
Loneliness	UCLA Loneliness Scale (short form) or comparable validated scale	Must be licensed/validated for Norwegian use
General skepticism	Adapted items on trust in unknown senders / online caution	New scale, pilot first
Phishing susceptibility	Scenario-based items + self-reported past experience	No real credential capture
Situational factors	Stress, workload, urgency sensitivity	Likert items
Protective behaviour	Verification habits, reporting, training recall	PMT-informed
Demographics	Age, study field, digital confidence	Controls

5.4 Interview guide (summary)

Interviews will explore:

- near misses or actual encounters with phishing;
- what the message looked like and why it felt credible;
- emotional state at the time (lonely, stressed, busy, afraid);
- whether they knew the advice but still hesitated or clicked;

- what would have helped them stop.

Full guide in Appendix A.

5.5 Analysis plan

- **Quantitative:** descriptive statistics, correlation, regression, mediation tests where assumptions hold.
- **Qualitative:** reflexive thematic analysis (Braun & Clarke, 2006).
- **Integration:** joint display of statistical patterns and lived experience quotes.

5.6 Ethics

- Application to NSD / institution ethics board before data collection.
- Informed consent, right to withdraw, anonymisation.
- Special care when asking about fraud victimisation or shame.
- No deceptive live phishing against participants without explicit ethical approval.

5.7 Limitations (anticipated)

- Self-report bias and social desirability.
- Convenience sample may overrepresent cybersecurity-aware students.
- Loneliness scales measure subjective experience, not objective isolation.
- Causality cannot be fully established in cross-sectional survey data.
- Wen et al. (2024) found weaker loneliness-fraud effects in younger adults; H1 may need nuance or subgroup analysis.

5.8 Pilot study plan

Before main data collection, a pilot with 10-15 OsloMet students/staff will test: (a) completion time (target under 12 minutes), (b) clarity of scenario vignettes, (c) floor/ceiling effects on loneliness items, and (d) whether skepticism items discriminate. Pilot feedback will refine wording and interview prompts.

5.9 Data management

- Survey hosted on institution-approved platform (e.g. Nettskjema / Qualtrics).
- Data stored encrypted on OsloMet-approved storage; access limited to researcher and supervisor.
- Retention aligned with NSD approval; deletion after thesis archive period.
- Interview audio transcribed; direct identifiers removed before analysis.

6. Expected Discussion and Conclusion

6.1 Expected findings

The thesis may find that:

- participants report high awareness but uneven safe behaviour;
- loneliness and stress increase openness to socially meaningful lures;
- skepticism toward strangers does not protect against authority or belonging cues;
- verification culture and non-shaming reporting matter as much as training content.

6.2 Practical implications

- Awareness campaigns should address emotional manipulation, not only fake links.
- Universities should normalise "pause and verify" without shame.
- Security training should include stress, loneliness, and social pressure scenarios.
- Design and process should make the safe action easier than the risky one.

6.3 Conclusion direction

Phishing is best understood as an attack on human meaning, not merely human ignorance. In a more skeptical yet often lonelier digital environment, defenders must protect not only systems but the moments when people seek connection, certainty, and control.

Appendix A: Interview Guide (draft)

1. Can you tell me about a time you received a suspicious message online - whether you clicked or not?
2. What made it seem believable or urgent?
3. How were you feeling that day (busy, stressed, lonely, distracted)?
4. Did the message feel like it came from a stranger, an institution, or someone you knew?
5. Did you know the security advice in that moment? If yes, why was it hard to follow?
6. Did you tell anyone? Why or why not?
7. What would help you stop faster next time?
8. Do you think loneliness or social pressure online affects how people respond to such messages?

Appendix B: Survey Questions (draft)

Section 1 - Background

- Age group
- Role (student / employee / other)
- Hours per day online for study/work
- Self-rated digital confidence (1-7)

Section 2 - Loneliness and social context

- UCLA-3 or equivalent validated items
- "I often feel that I lack companionship" (Likert)
- "I feel in tune with the people around me" (reverse)

Section 3 - Skepticism and awareness

- "I am generally skeptical of messages from unknown senders."
- "I feel confident that I can recognise phishing."
- "I have received cybersecurity training in the last 12 months."

Section 4 - Scenario susceptibility

Present 6 short vignettes. For each: "How likely would you be to click, reply, or follow instructions?" (1-7 Likert). See Appendix C for full vignette text.

Section 5 - Protective behaviour

- "I verify unexpected requests through a separate channel."
- "I would report a suspicious message even if I almost fell for it."
- "I worry about being judged if I report a near miss."

Appendix C: Phishing Scenarios (draft)

Each vignette is followed by a 1-7 likelihood scale and optional open comment. Scenarios are fictional and contain no live links.

#	Lure type	Scenario text	Construct tested
S1	Authority / safety	You receive an email from "OsloMet IT Security" saying your student account will be locked in 2 hours unless you verify your password via a link. You are preparing for an exam submission deadline today.	Urgency + institutional trust
S2	Belonging / social	A person you met briefly at a student event messages you on Instagram: "Hey, we should grab coffee - here's a link to my calendar." You have few friends on campus and have felt lonely lately.	Social relevance + loneliness
S3	Order / control	An SMS says a package is waiting and you must pay a 39 NOK customs fee via a link to avoid return. You recently ordered something online but are not sure which store.	Uncertainty + small cost
S4	Status / opportunity	LinkedIn message: "We saw your profile and want to invite you to a paid research assistant role. Complete this short form today - limited spots." You are job hunting.	Ambition + scarcity
S5	Help / compliance	Phone call from someone claiming to be bank support: "We detected suspicious activity. Stay on the line and approve the notification on your phone so we can secure your account."	Authority + fear
S6	Peer impersonation	Email from a classmate's address: "Can you quickly buy these gift cards for our group project? I'll pay you back tonight." The tone matches how they usually write.	Known sender + helpfulness

Analysis note: S2 and S6 test social/belonging hypotheses; S1, S3, S5 test authority and control; S4 tests status. Compare responses by loneliness quartile and skepticism score.

Appendix D: Ethics and NSD Application (draft)

Project title

Phishing Between Loneliness and Skepticism: Human Vulnerability to Social Engineering in Norwegian Higher Education

Data collected

- Anonymous survey responses (quantitative + optional open text)
- Semi-structured interviews (audio recorded with consent, pseudonymised)
- No collection of real credentials, bank details, or live phishing deployment

Legal basis and consent

Participation is voluntary. Informed consent form explains purpose, data use, storage period, right to withdraw (including deletion of interview data before analysis lock), and contact details. Participants must be 18+.

Sensitive topics

Questions about loneliness, fraud victimisation, and shame may cause discomfort. Information sheet will include links to student welfare (SiO/Samskipnaden), Mental Helses hjelpetelefon, and NSM reporting channels. No question requires disclosure of financial loss; "prefer not to say" options included.

Risk assessment

- **Participant risk:** Low. No deceptive live attacks. Scenario vignettes are clearly fictional.
- **Privacy risk:** Medium-low. Interview quotes require careful anonymisation (remove programme, names, dates).
- **Benefit:** Improved understanding of student phishing vulnerability; recommendations for HE security culture.

NSD checklist (pre-submission)

1. Finalise survey instrument and consent forms
2. Register project in NSD/REK portal
3. Obtain OsloMet faculty ethics approval if required
4. Document data processor agreements for survey platform

Appendix E: Timeline (proposed)

Phase	Period	Activities	Deliverable
1. Alignment	Sep-Oct 2026	Supervisor meetings, refine RQ and hypotheses, literature in Zotero	Approved problem statement
2. Instrument design	Nov-Dec 2026	Validate scales, translate/adapt items, build survey	Survey v1.0 + interview guide
3. Ethics	Jan 2027	NSD application, consent forms, pilot ethics check	NSD approval
4. Pilot	Feb 2027	10-15 respondents, revise instruments	Survey v1.1
5. Main survey	Mar-Apr 2027	Recruit 120-180 via OsloMet channels	Quantitative dataset
6. Interviews	Apr-May 2027	12-18 interviews, transcription	Qualitative corpus
7. Analysis	Jun-Jul 2027	Stats + thematic analysis, integration	Results chapter draft
8. Writing	Aug-Nov 2027	Full thesis draft, supervisor feedback loops	Thesis draft
9. Submission	Dec 2027	Final edit, formatting, submission	Master thesis

Adjust dates to match OsloMet MSc Cybersecurity programme deadlines.

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Note: Reference list to be expanded and verified in Zotero before submission. Some items require full bibliographic cleanup.

Next Steps (work plan)

1. Book supervisor meeting with this draft and core hypothesis.
2. Confirm population: OsloMet students/staff only or national HE sample.
3. Import references into Zotero and verify APA formatting.
4. Validate loneliness (UCLA-3) and draft skepticism scale for Norwegian context.
5. Submit NSD ethics application (see Appendix D).
6. Pilot survey with 10-15 respondents (Feb 2027 per timeline).
7. Revise hypotheses after pilot; pre-register analysis plan if required.
8. Begin main data collection spring 2027.

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